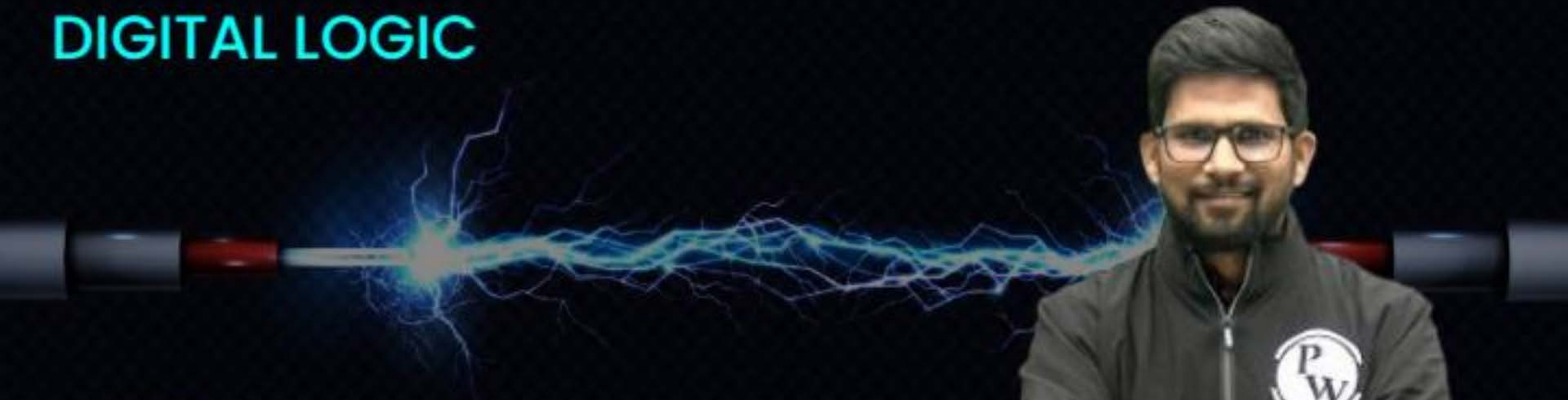
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 06

Miscellaneous Topics



By- Chandan Gupta Sir

Recap of Previous Lecture

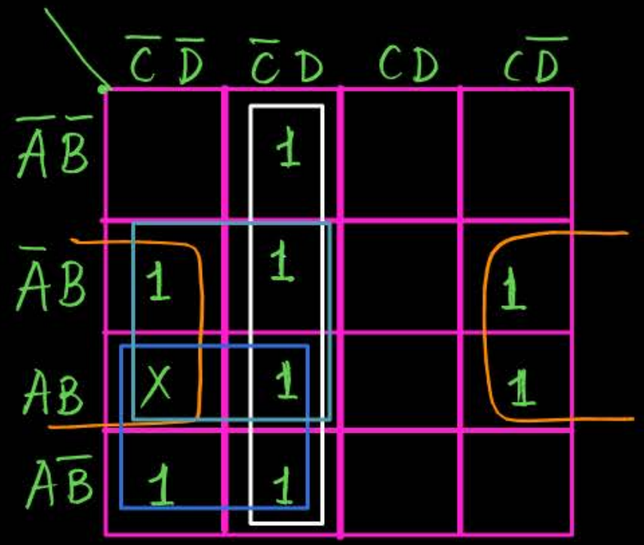


- PI, EPI
- Question Discussion on Number System
- PI, EPIs



- State transition Diagram
- Parity generator & checker

#Q.



$PI_s \longrightarrow \bar{C}D,\ B\bar{D},\ A\bar{C},\ B\bar{C}$

$EPI_s \longrightarrow \bar{C}D,\ B\bar{D},\ A\bar{C}$

#Q.

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1		X	1
$\bar{A}B$	1	1	1	1
AB				X
$A\bar{B}$	1			1

PIs →

EPIs →

H.W.

#Q.

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1		1	1
$\bar{A}B$	X			
AB	1		1	X
$A\bar{B}$	1		1	1

H.W.

PIs ⟶

EPIs ⟶

#Q.

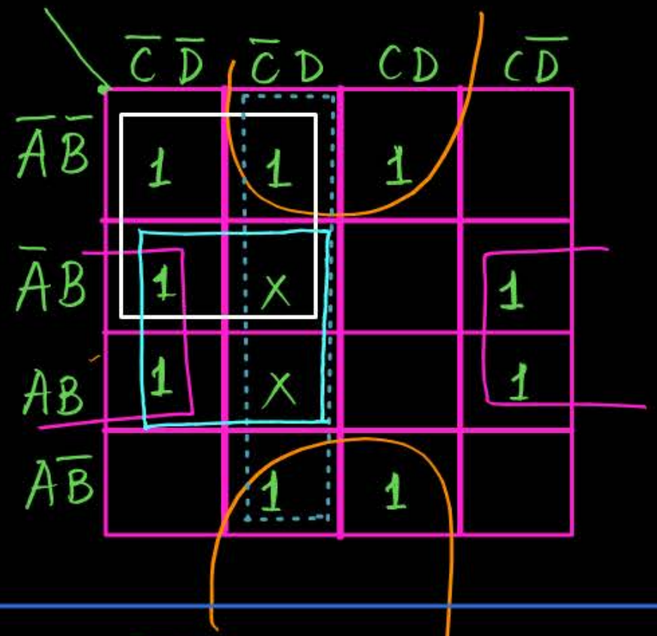
	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1	1		
$\bar{A}B$	1	1		
AB	X	1	1	1
$A\bar{B}$	X			

PIs →

EPIs →

H.W.

#Q.



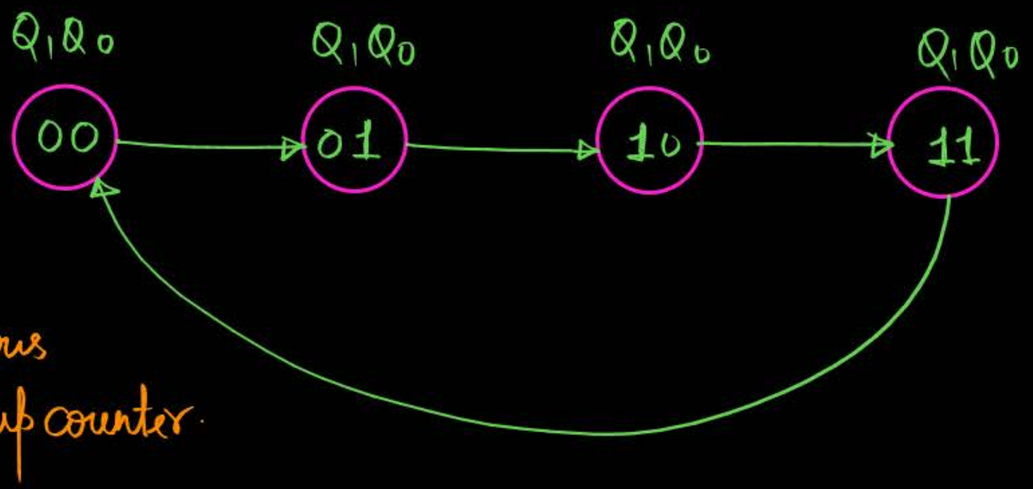
$PI_s \rightarrow \bar{A}\bar{C}, B\bar{D}, \bar{B}D, \bar{C}D, B\bar{C}$

$EPI_s \rightarrow \bar{A}\bar{C}, B\bar{D}, \bar{B}D,$

State Transition Diagram

What is state transition diagram ?

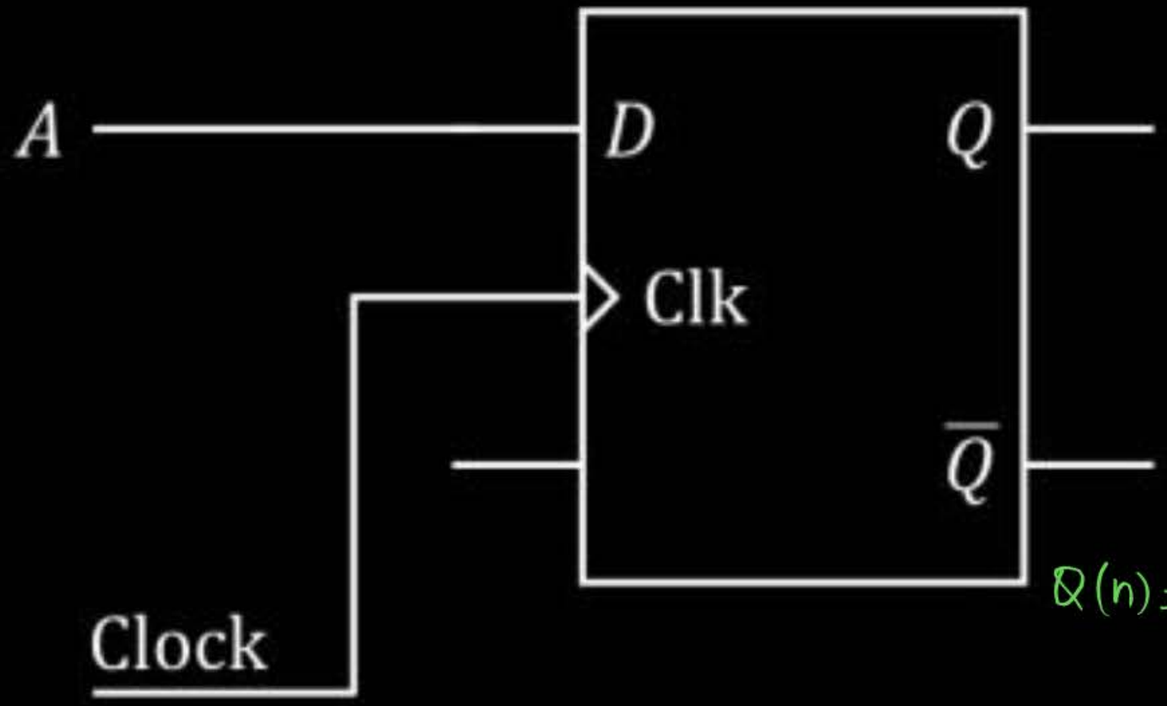
State diagram is a diagram that represents, all the possible transitions of a sequential ckt.



$$Q_1(n+1) = Q_0(n) \oplus Q_1(n)$$
$$Q_0(n+1) = \overline{Q_0}(n)$$

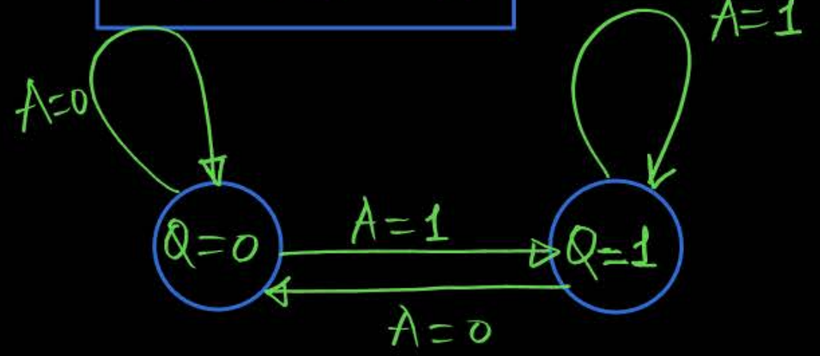
Synchronous
2-bit up counter.

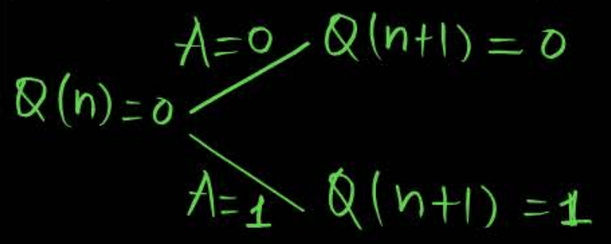
- Lets understand it with an example :



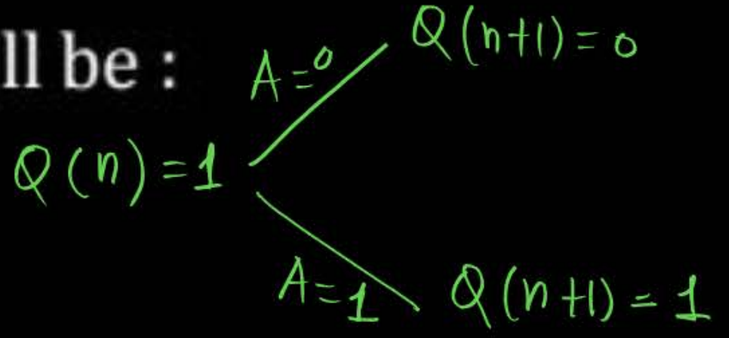
$Q(n+1) = D$

$Q(n+1) = A$



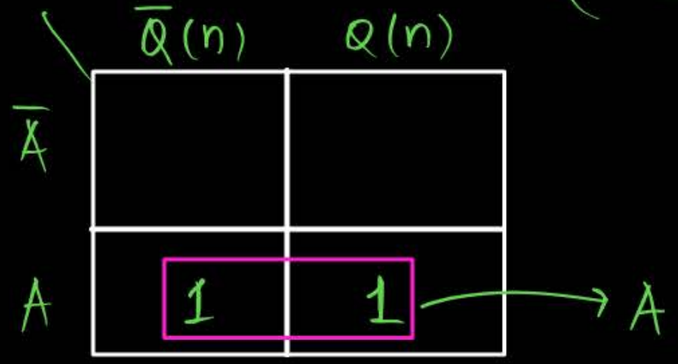


State transition diagram for above circuit will be :



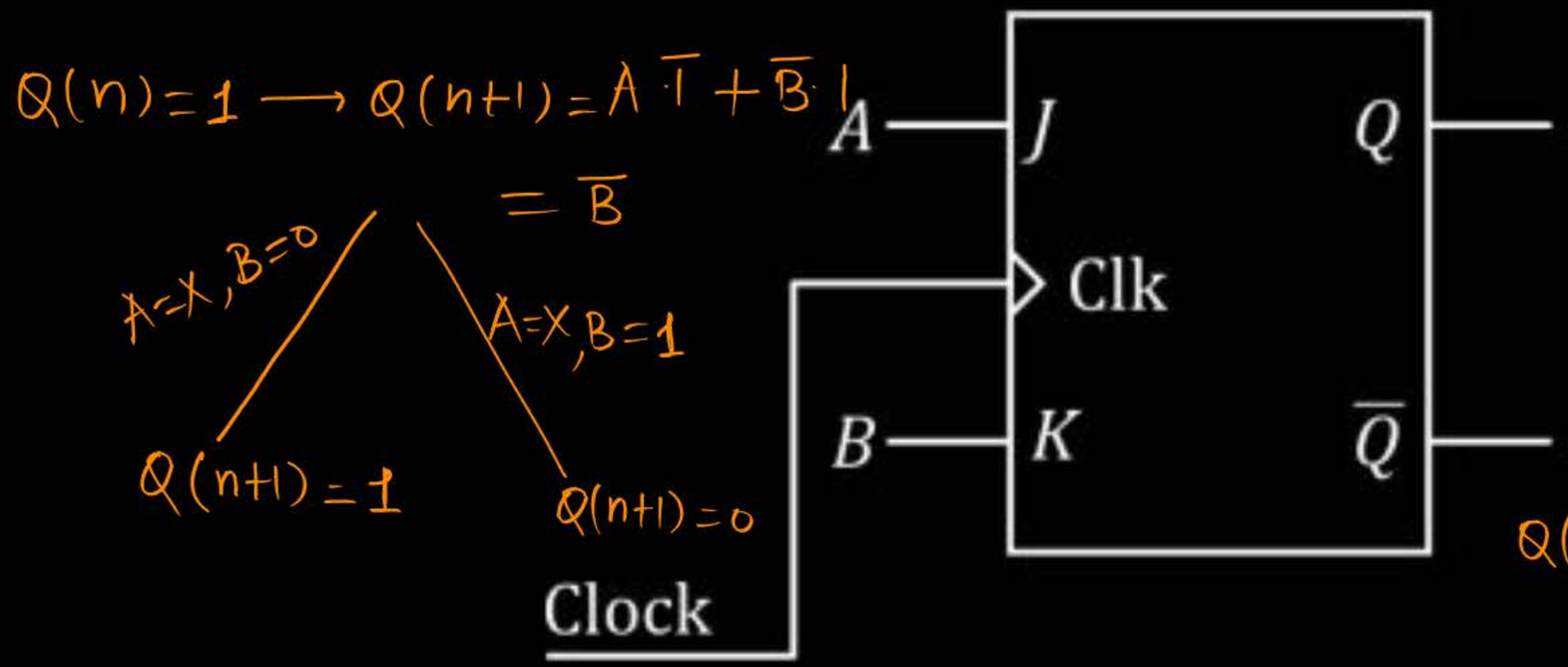
A	Q(n)	Q(n+1)
0	0	0
0	1	0
1	0	1
1	1	1

$Q(n+1)[A, Q(n)] = \Sigma(2,3) = A$



$Q(n+1) = A$

- A sequential circuit is as given below :



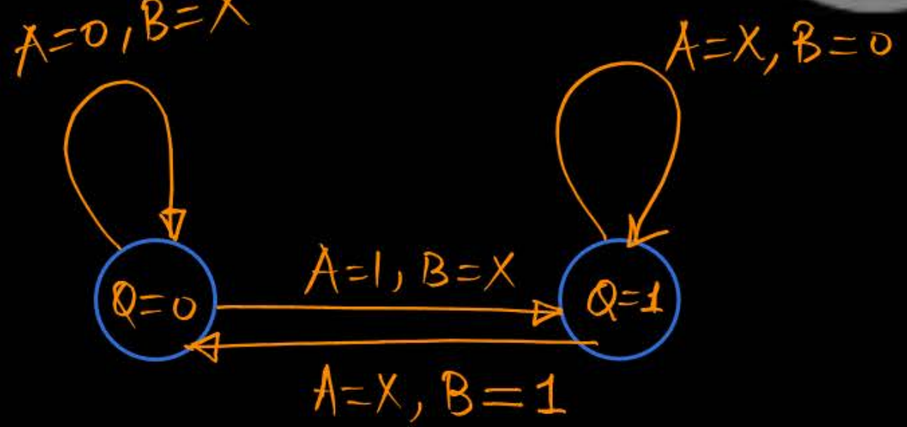
$Q(n)=1 \longrightarrow Q(n+1)=A\cdot\overline{1}+\overline{B}\cdot 1$

$=\overline{B}$

$A=X, B=0$ → $Q(n+1)=1$

$A=X, B=1$ → $Q(n+1)=0$

$Q(n+1)=A\overline{Q}(n)+\overline{B}Q(n)$



$Q(n)=0 \longrightarrow Q(n+1)=A+\overline{B}\cdot 0=A$

$A=0, B=X$ → $Q(n+1)=0$

$A=1, B=X$ → $Q(n+1)=1$

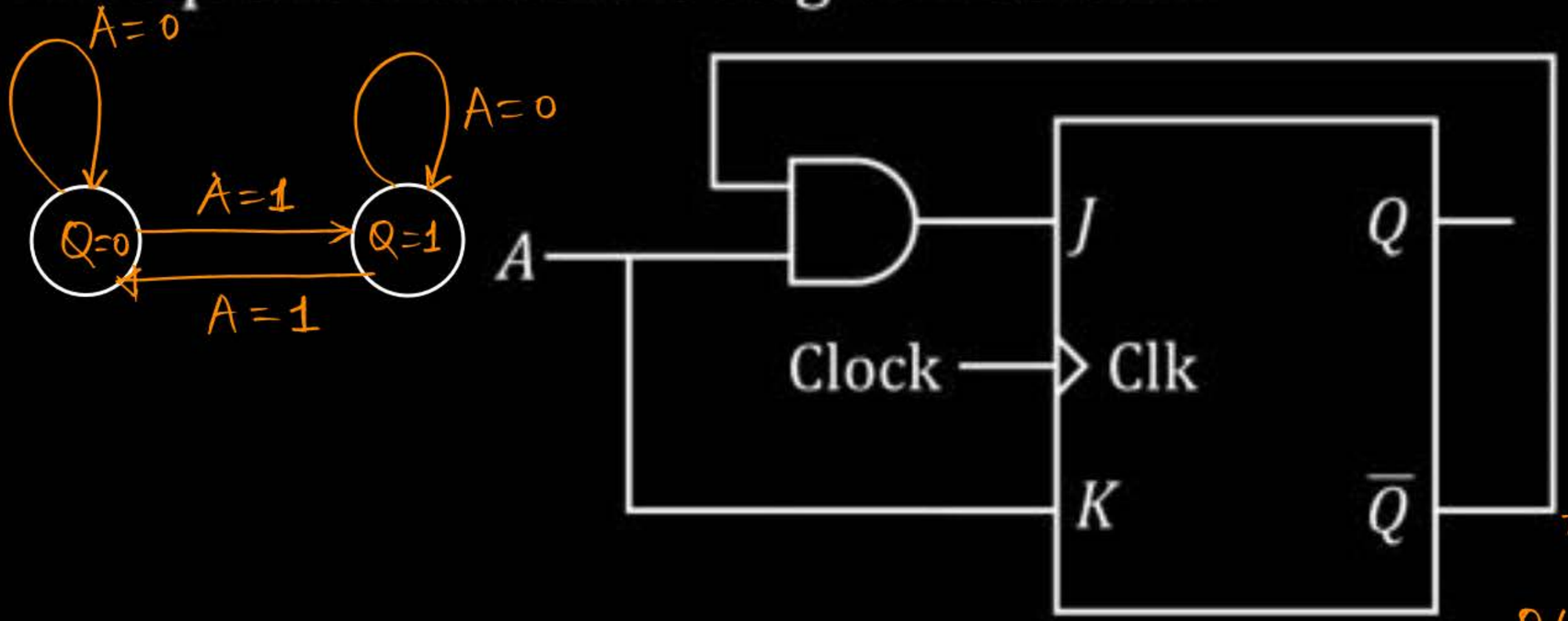
State transition diagram of above circuit will be :

A B Q(n) Q(n+1)

H.W.

Question

A sequential circuit is as given below :

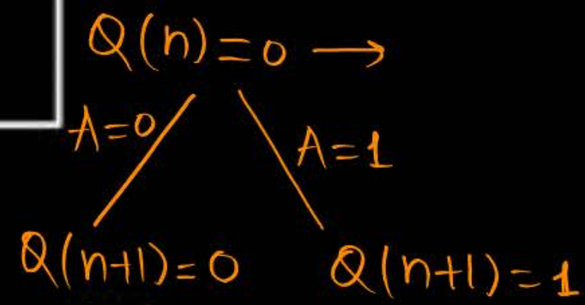


$$Q(n+1) = J\overline{Q}(n) + \overline{K}Q(n)$$

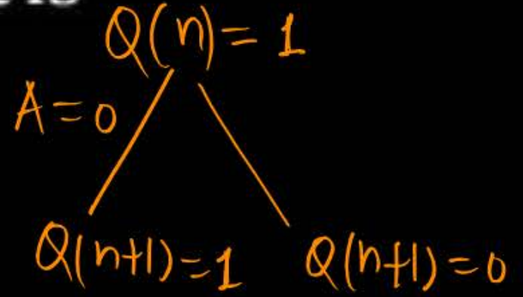
$$= A\overline{Q}(n) \cdot \overline{Q}(n) + \overline{A}Q(n)$$

$$= A\overline{Q}(n) + \overline{A}Q(n)$$

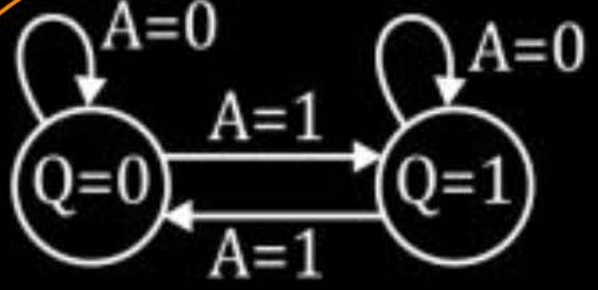
$$Q(n+1) = A \oplus Q(n)$$



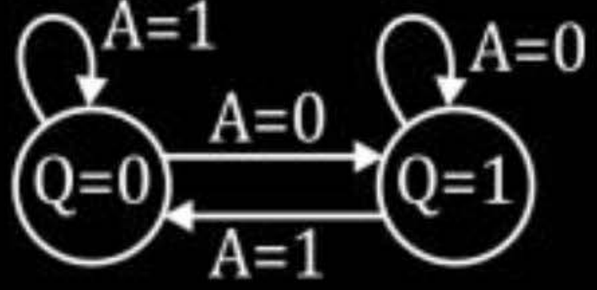
State transition diagram of above sequential circuit is



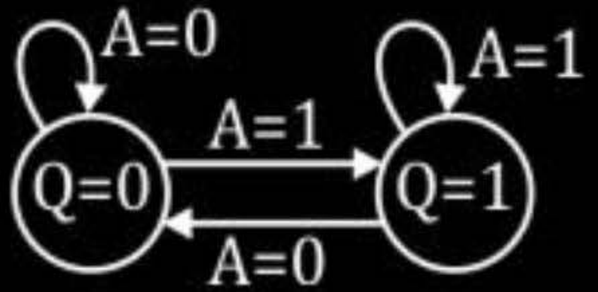
(a)



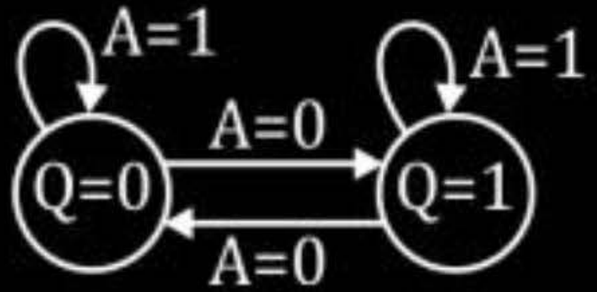
(b)



(c)

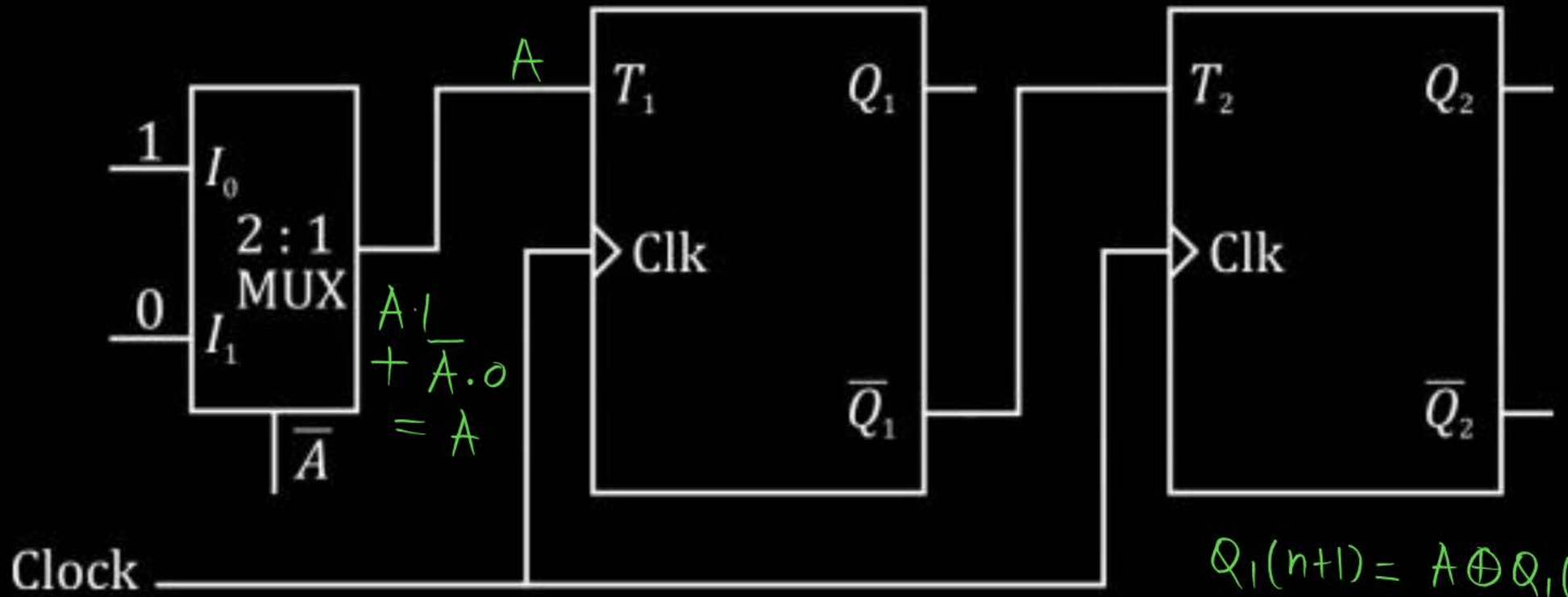


(d)



Question

A sequential circuit is as given below :



$Q_1(n+1) = A \oplus Q_1(n)$

$Q_2(n+1) = \overline{Q}_1(n) \oplus Q_2(n)$

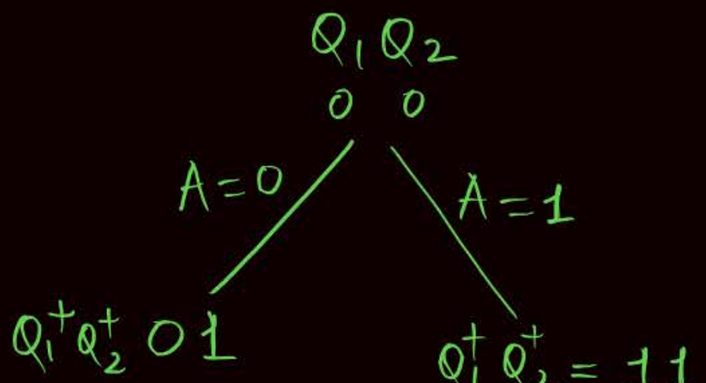
$= Q_1(n) \odot Q_2(n)$

State transition diagram of above circuit will be :

$Q_1(n)\,Q_2(n) = 0\,0$

$Q_1(n+1) = A \oplus 0 = A$

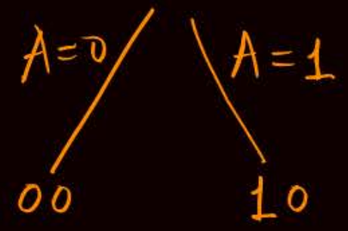
$Q_2(n+1) = 0 \odot 0 = 1$



$Q_1(n)\,Q_2(n) = 0\,1$

$Q_1(n+1) = 0 \oplus A = A$

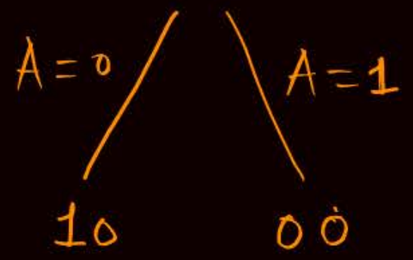
$Q_2(n+1) = 0$



$Q_1(n)\,Q_2(n) = 1\,0$

$Q_1(n+1) = A \oplus 1 = \overline{A}$

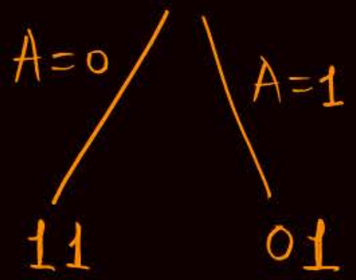
$Q_2(n+1) = 0$



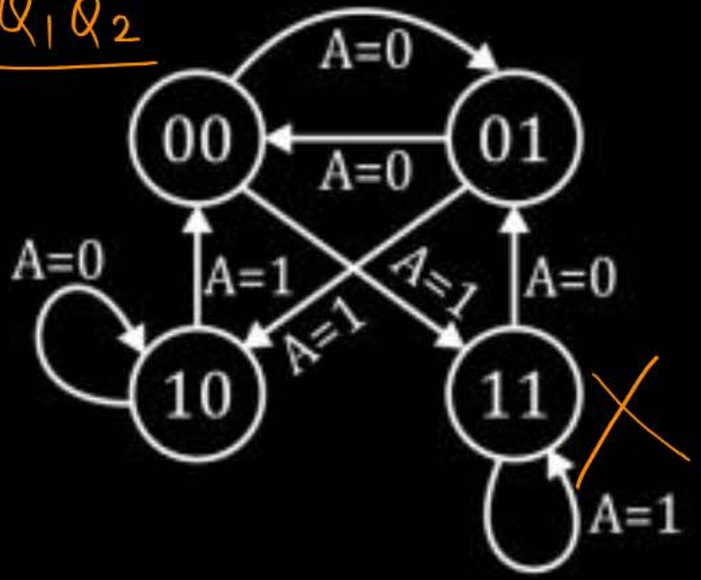
$Q_1(n)\,Q_2(n) = 1\,1$

$Q_1(n+1) = A \oplus 1 = \overline{A}$

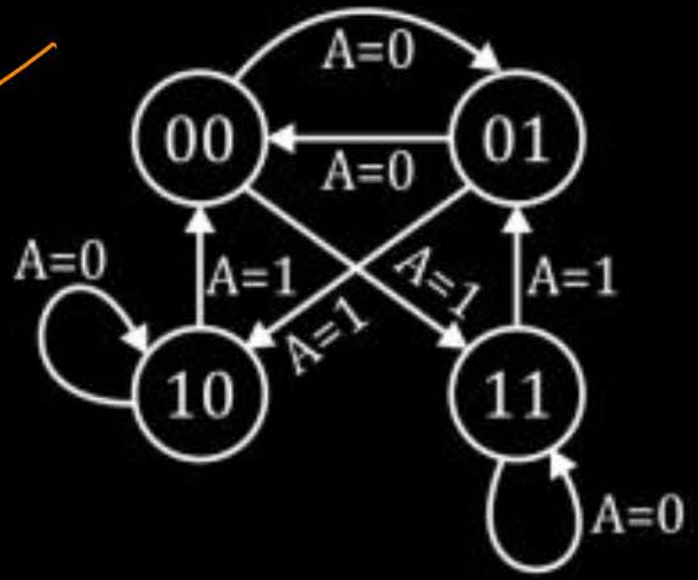
$Q_2(n+1) = 1$



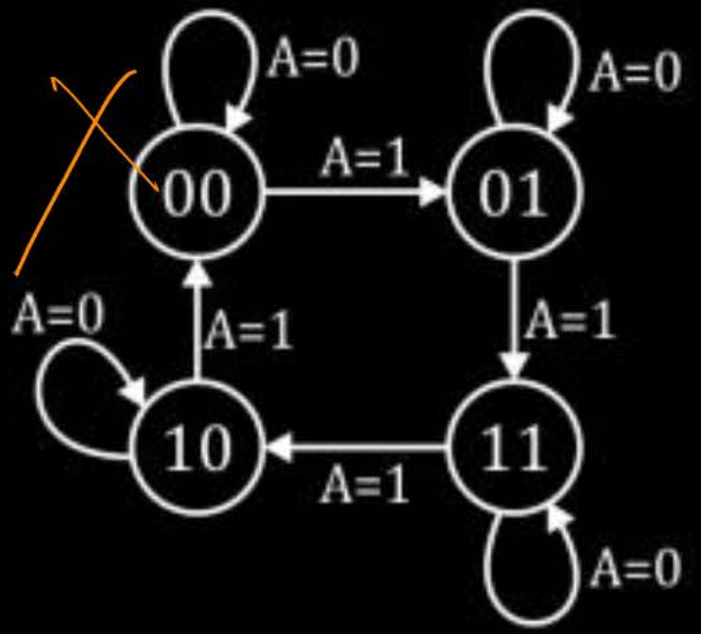
(a)


Q_1Q_2
A=0
00
01
A=0
A=0
A=1
A=1
A=1
A=0
10
11
A=1

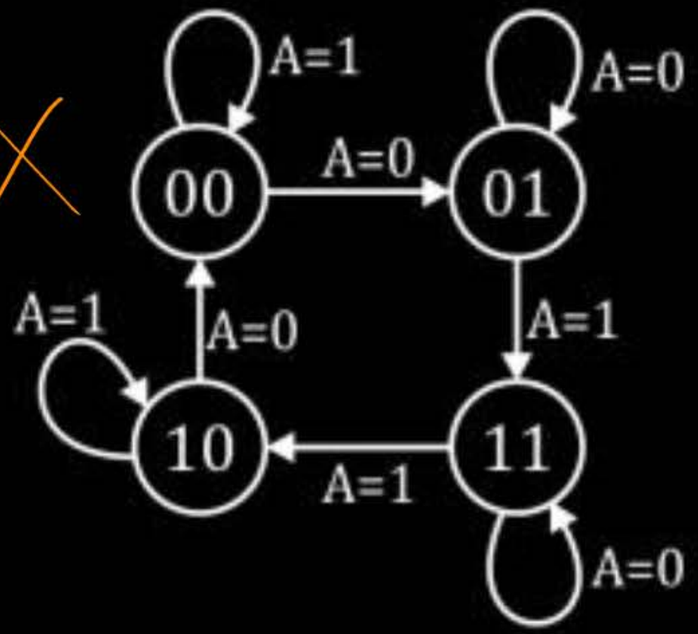
(b)


A=0
00
01
A=0
A=0
A=1
A=1
A=1
A=1
10
11
A=0

(c)

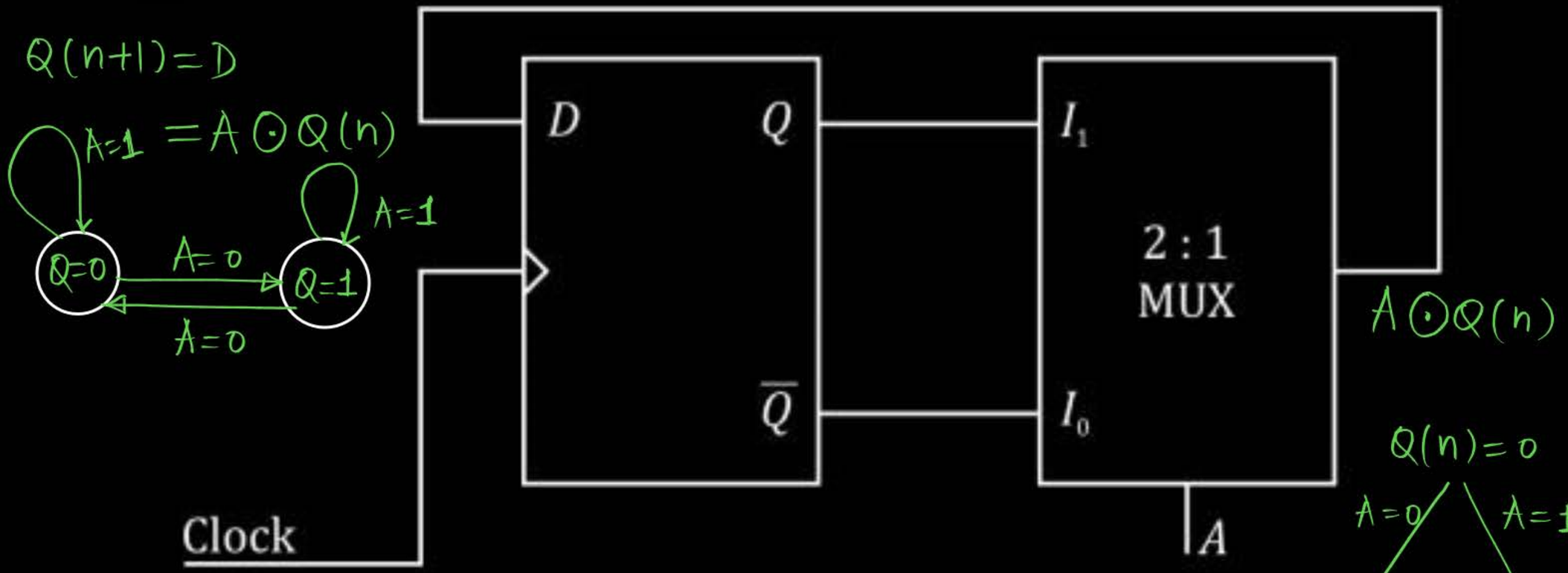

A=0
A=0
00
A=1
01
A=0
A=1
A=1
10
11
A=1
A=0

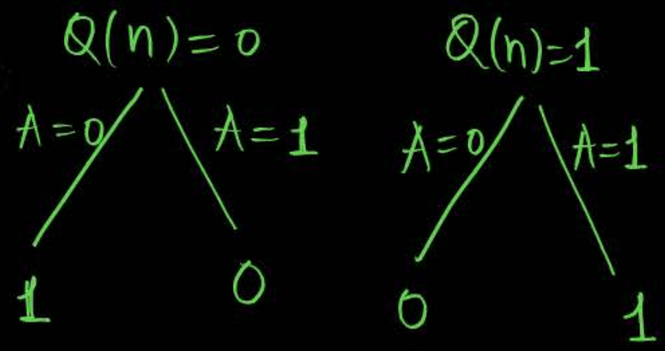
(d)


A=1
A=0
00
A=0
01
A=1
A=0
A=1
10
11
A=1
A=0

Question

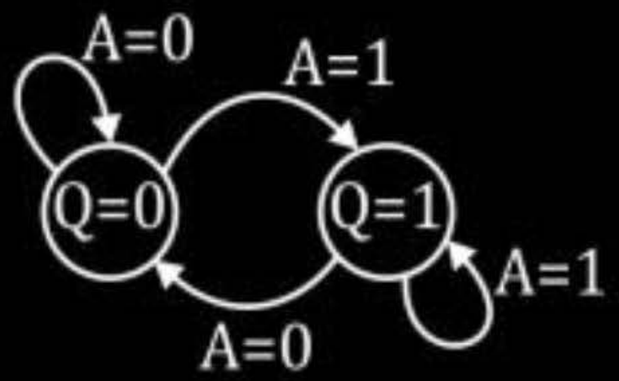
A sequential circuit is as given below :



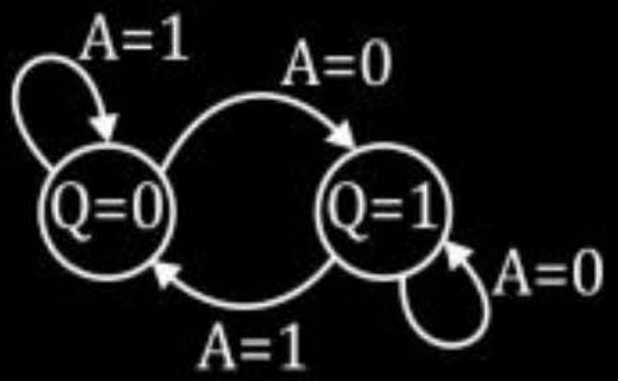


State transition diagram of above circuit will be :

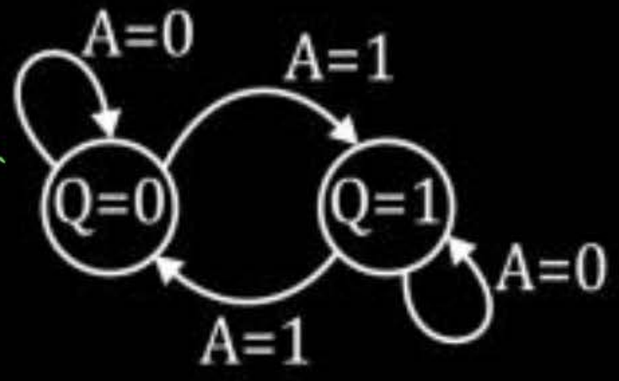
(a)



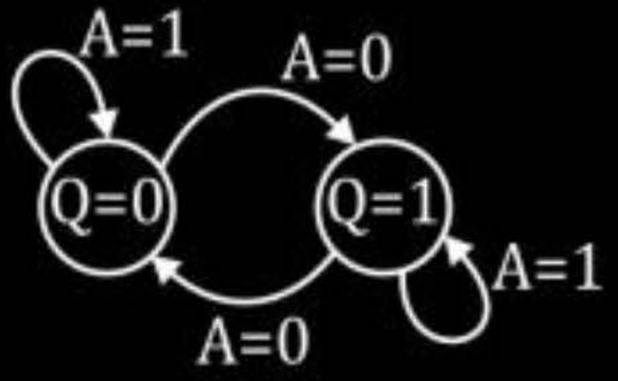
(b)



(c)

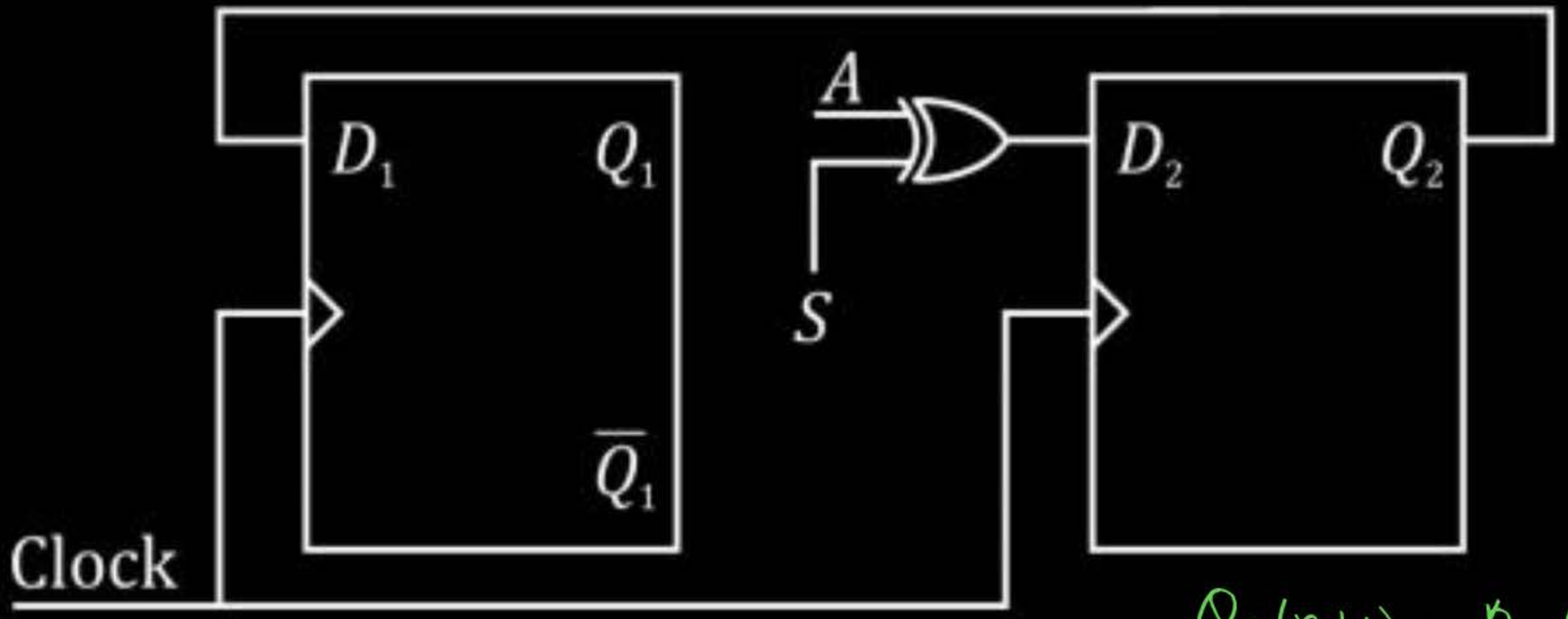


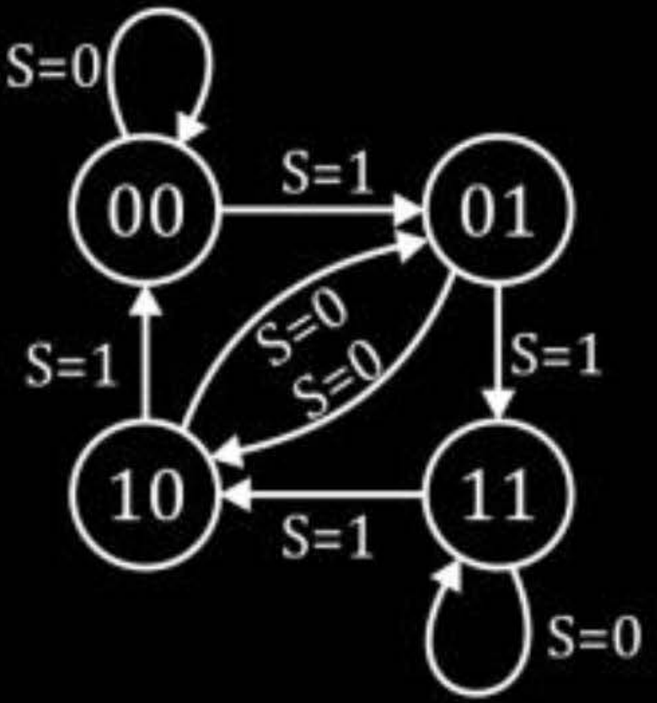
(d)



Question

The digital logic shown in the figure satisfies the given state diagram when Q_1 is connected to input A of the XOR gate :





$Q_1(n+1) = D_1 = Q_2(n)$

$Q_2(n+1) = D_2 = S \oplus Q_1(n)$

$Q_1(n+1) = Q_2(n)$

$Q_2(n+1) = A \odot S$

Suppose XOR gate is replaced by XNOR gate then which of the option preserves the state diagram :

(a) Input A is connected to $\bar{Q}_2$

(b) Input A is connected to Q_2

(c) Input A is connected to $\bar{Q}_1$ and S is complimented.

(d) Input A is connected to $\bar{Q}_1$

$\bar{Q}_1$ or S is complemented.

[MCQ]

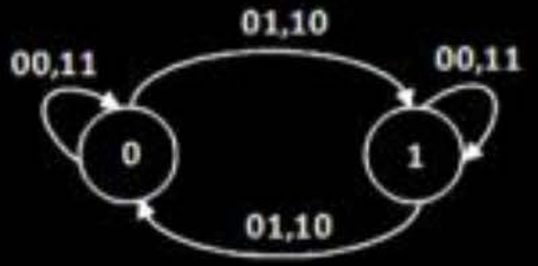
Consider the following state diagram and its realization by a JK flip flop :
The combinational circuit generates J and K in terms of x, y and Q. The Boolean expressions for J and K are :

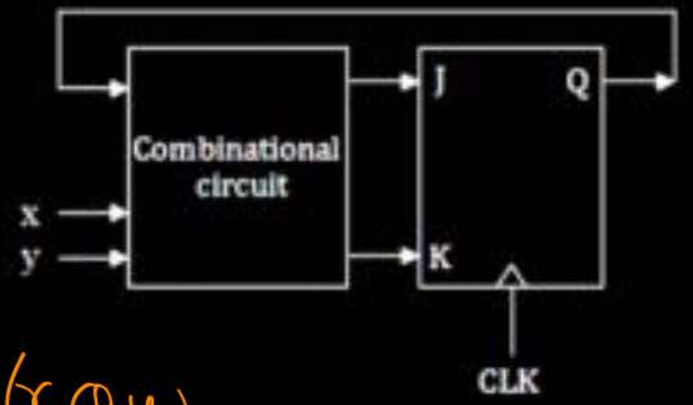
A. $(x \oplus y)'$ and $(x \oplus y)'$

B. $(x \oplus y)'$ and $x \oplus y$

C. $x \oplus y$ and $(x \oplus y)'$

D. $x \oplus y$ and $x \oplus y$





$Q(n+1) = J\bar{Q}(n) + \bar{K}Q(n)$

$x, y = 00, 11$

$J = 0, K = 0$

$J = (x \oplus y)$

$K = x \oplus y$

x	y	Q(n)	Q(n+1)
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

$$Q(n+1)\,[x, y, Q(n)] = \Sigma(1, 2, 4, 7)$$

$$Q(n+1) = [x \oplus y] \oplus Q(n)$$

$J = \bar{A}$, $K = A$ $\longrightarrow$ $Q(n+1) = \bar{A} \oplus Q(n)$

Parity generator & Parity checker.

A	B	C	P_e	P_o
0	0	0	0	1
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	0

$$P_e(A,B,C) = A \oplus B \oplus C$$

$$P_e(A,B,C,D) = A \oplus B \oplus C \oplus D$$

$$P_o(A,B,C) = \overline{A \oplus B \oplus C}$$

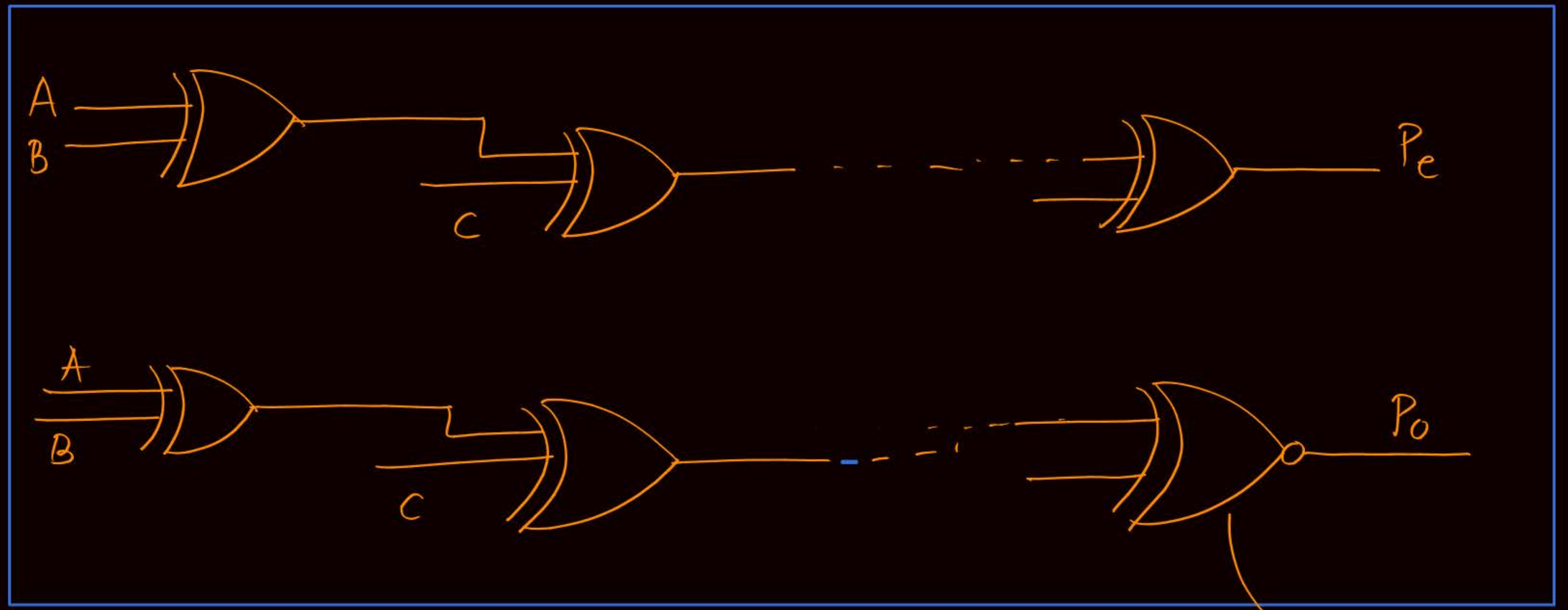
$$P_o(A,B,C,D) = \overline{A \oplus B \oplus C \oplus D}$$

$$P_o(A,B,C) = A \oplus B \odot C$$

$$P_o(A,B,C,D) = \overline{A \oplus B \oplus C \odot D}$$

$$P_o(A,B,C,D,E) = \overline{A \oplus B \oplus C \oplus D \oplus E}$$

$$= A \oplus B \oplus C \oplus D \odot E$$



→ only last gate is 2 i/p XNOR other are XOR.

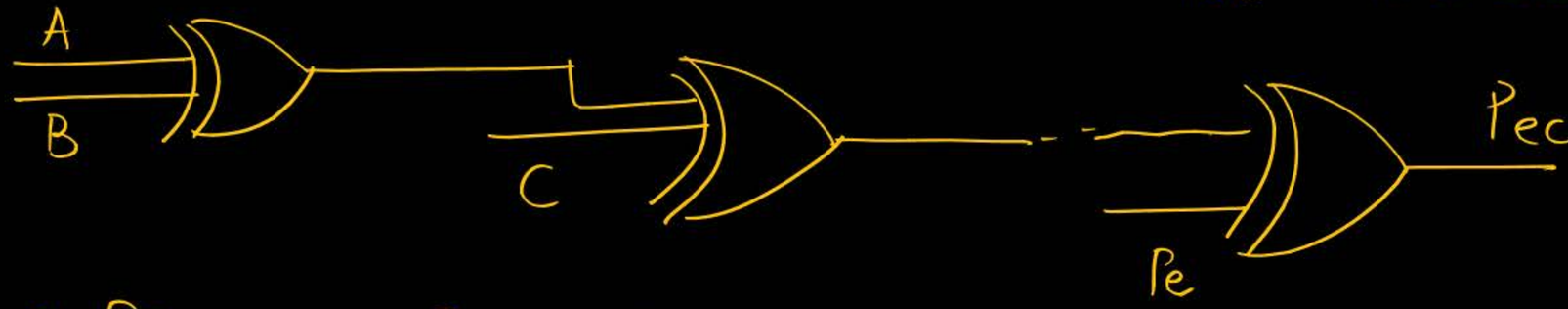
Parity Checker: Even Parity checker.

A B C Pe Pec

$P_{ec} = 1 \rightarrow$ there is error

$= 0 \rightarrow$ No error

$P_{ec} = A \oplus B \oplus C \oplus P_e$



$P_{ec} = 1 \rightarrow$ then error

$= 0 \rightarrow$ then no error

$P_{ec}[A, B, C, D, P_e] = A \oplus B \oplus C \oplus D \oplus P_e$

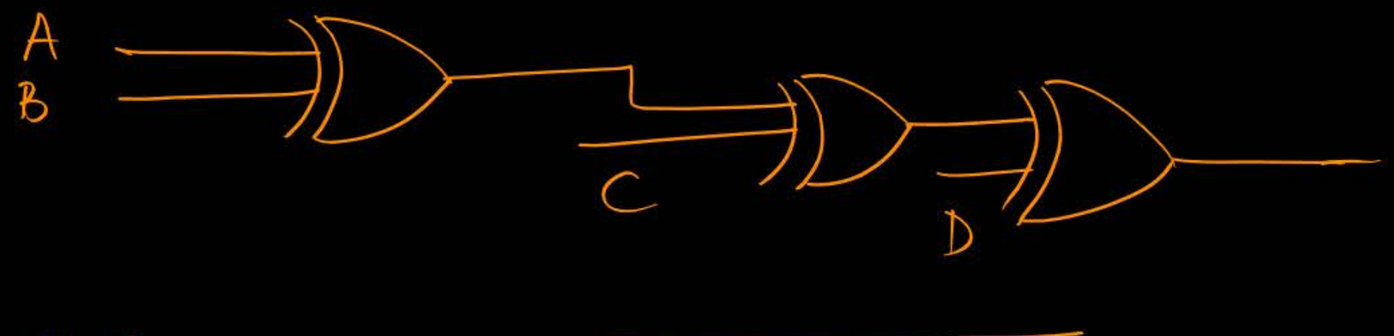
Odd Parity checker

A B C P_o P_{oc}

$$P_{oc}[A, B, C, P_o] = \overline{A \oplus B \oplus C \oplus P_o}$$

$P_{oc} \rightarrow 1 \rightarrow$ there is error

$0 \longrightarrow$ no Error



$P_{oc} \rightarrow 1 \rightarrow$ Error

$\rightarrow 0 \rightarrow$ No Error.

$$P_{oc}[A, B, C, D, P_o] = \overline{A \oplus B \oplus C \oplus D \oplus P_o}$$

Functionally Complete Function

- **Total-0 preserving function** (T_0)

$f(A, B, C)$ → if $f(0,0,0) = 0$ → said to be total-0 preserving function T_0

if $= 1$ → It is not a total-0 preserving function

Not T_0

$f(A, B, \bar{C}) = AB + \bar{C}$ → $f(A, B, D) = AB + D$

$f(0,0,0) = 0 + 0 = 0$ → T_0

$A = 0$

$B = 0$

$\bar{C} = 0 \rightarrow C = 1$

Total-1 preserving function.

$f(A,B,C) \rightarrow$ if $f(1,1,1) = 1 \longrightarrow$ Total-1-preserving function T_1

if $f(1,1,1) = 0 \longrightarrow$ It is not a total-1 preserving function.
Not T_1

- $f(A,B,\overline{C}) = AB + \overline{C}$

$f(1,1,1) = 1 \cdot 1 + 1 = 1 \longrightarrow T_1$

$f(A,B,\overline{C}) = A\overline{B} + C$

$f(1,1,1) = 1 \cdot \overline{1} + 0 = 0 \longrightarrow$ Not T_1

Linear Function:

A function is said to be linear if for all even no. 1's combination O/P is '1' and for all odd no. of 1's combinations O/P is '0' and vice-versa.

f(A,B)	A	B	NL OR	NL AND	NL NAND	NL NOR	L XOR	L XNOR
	0	0	0	0	1	1	0	1
	0	1	1	0	1	0	1	0
	1	0	1	0	1	0	1	0
	1	1	1	1	0	0	0	1

- Compliment of a linear function will always be linear function.

Monotonic function

- If one of the i/p changes from 0—1 and O/p is non decreasing then function is said to be Monotonic function.

$f(A, B)$

A	B	M OR	M AND	NM NAND	NM NOR	NM XOR	NM XNOR
0	0	0	0	1	1	0	1
0	1	1	0	1	0	1	0
1	0	1	0	1	0	1	0
1	1	1	1	0	0	0	1

H.W

Q.1. $f(A,B,C) = \overline{A} + \overline{B}C$, T_0, T_1, L, M

Q2. $f(A,\overline{B}) = \overline{A}B$, T_0, T_1, L, M.

Q.3. $f(\overline{A},B) = A + \overline{B}$, T_0, T_1, L, M.

Q.4 $f(\overline{A},\overline{B}) = \overline{A}B$, T_0, T_1, L, M.

Q.5. $f(A,\overline{B},C) = \overline{A}B + \overline{C}$, T_0, T_1, L, M.

$f(A,\overline{B}) = \overline{A} + \overline{B}$

$\overline{B} = D$

$f(A,D) = \overline{A} + D$

$= \Sigma(0,1,3)$

Not T_0, T_1, NL, NM.

00	—	1
01	→	1
10	—	0
11	—	1



Topic : 2 Min Summary

- State transition Diagram
- Type of function.

Thank you

GW

Soldiers !

